

TSENAT: Tsallis Entropy Analysis Toolbox

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Overview

TSENAT (Tsallis Entropy Analysis Toolbox) allows the modelization and quantification of isoform-usage complexity in RNA-seq data using Tsallis entropy, a scale-dependent diversity measure distinct from abundance-focused tools (DESeq2, edgeR) and differential transcript usage packages (DRIMSeq). By tuning the entropy index parameter (q), TSENAT enables to examine transcriptome heterogeneity at different scales: rare variants (low q) or dominant isoforms (high q). This package enables computing Tsallis entropy and Tsallis divergence from transcript-level abundance estimates, comparing measures between groups, and visualizing scale-dependent differences via q -curves.

Motivation and Bioconductor Contribution

Common RNA-seq tools focus on either total gene abundance changes or individual transcript usage shifts, yet genes often remodel their isoform landscapes without changing overall expression—a phenomenon missed by these approaches. TSENAT makes possible to quantify this isoform-usage diversity directly via Tsallis entropy, allowing to capture biological signals invisible to abundance- or proportion-based summaries.

In this guide, we demonstrate the complete workflow: preprocessing transcript counts, computing entropy across entropic indices, testing for between-group differences, and visualizing scale-dependent complexity.

High-level workflow

1. **Load and preprocess** transcript counts and filter low-abundance transcripts.
2. **Compute diversity** (Tsallis entropy) and **divergence** (Tsallis divergence).
3. **Test for differences** in entropy patterns between groups across entropic indices.
4. **Visualize results**.

Design assumptions: This guide assumes paired or longitudinal designs with ≥ 6 -8 samples per group for adequate power to detect entropy shifts while controlling false discovery rate. Smaller sample sizes may be underpowered to detect subtle isoform complexity changes.

Installation

To install TSENAT from Bioconductor:

```
if (!require("BiocManager", quietly = TRUE))
  install.packages("BiocManager", repos = "http://cran.us.r-project.org")
BiocManager::install("TSENAT")
```

Then load the package:

```
library(TSENAT)
```

Quick Start

For users who want to see results quickly, here is a minimal workflow:

```
library(TSENAT)
library(SummarizedExperiment)

# Load example data
data("readcounts", package = "TSENAT")
readcounts <- as.matrix(readcounts)

# Load sample metadata and annotation
metadata_df <- read.table(
  system.file("extdata", "metadata.tsv", package = "TSENAT"),
  header = TRUE, sep = "\t"
)
gff3_file <- system.file("extdata", "annotation.gff3.gz", package = "TSENAT")

# Create analysis object with transcript counts, annotation, and metadata
config <- TSENAT_config(
  sample_col = "sample",
  condition_col = "condition",
  paired = TRUE,
  subject_col = "paired_samples",
  control = "normal",
  q = seq(0, 2, by = 0.05)
)

## Build TSENATAnalysis object
analysis <- build_analysis(
  readcounts = readcounts,
  tx2gene = gff3_file,
  metadata = metadata_df,
  config = config,
  tpm = tpm,
  effective_length = effective_length)

# Filter low-abundance transcripts
analysis <- filter_analysis(analysis)

# Compute Tsallis entropy using S4 wrapper (using single q value for quick start)
analysis <- calculate_diversity(analysis)

# Plot overall q-curve
p_qcurve <- plot_diversity_spectrum(
  analysis,
  dev_width = 12,
  dev_height = 8)
print(p_qcurve)
```

What is Entropy and Tsallis Entropy?

The concept of **entropy** originated in Claude Shannon’s landmark 1948 paper on information theory (Shannon 1948), which established that information content could be quantified mathematically through the fundamental concepts of uncertainty and surprise. Shannon entropy became the foundation for understanding complexity across mathematics, physics, and biology—if you draw a transcript from a distribution, how predictable is the outcome?

However, Shannon entropy treats all elements equally, regardless of their frequency: it weights rare and common elements identically. This led mathematicians and physicists to explore **generalized entropy families** that could emphasize different aspects of distributions. Two major generalizations emerged: Rényi’s parametric family of entropies and Tsallis entropy, both of which introduced tunable parameters that allow sensitivity to rare versus abundant elements.

The key insight is that Tsallis and Rényi entropies, despite appearing different mathematically, can be unified within a coherent framework through generalized logarithmic and exponential functions (Tsallis 2017). Both frameworks answer a fundamental question: **What organizational scales matter?** By changing the q parameter, you shift emphasis from rare (low q) to abundant (high q) elements—qualitatively different perspectives on the same distribution.

More recently, **Hill numbers** (Chao et al. 2010) provided a modern ecological framework that reinterprets all generalized entropy measures as “true diversity” of different orders. This formulation clarified that diversity questions have scale-dependent answers: rare species and dominant species reveal different ecological (or in our case, transcriptomic) truths. The Hill numbers framework unified richness ($q=0$), Shannon entropy ($q=1$), Simpson/Gini ($q=2$), and higher-order generalizations under a single mathematical umbrella.

TSENAT applies this intellectual progression to RNA-seq data: instead of asking only “which genes change abundance,” it asks “how does the organization of isoforms change across multiple scales of complexity?” By leveraging Tsallis entropy’s multi-scale nature and the Hill numbers interpretation, TSENAT captures the full diversity spectrum of isoform organization, revealing biological signals invisible to traditional abundance-focused methods.

Why This Matters for RNA-seq: The Isoform Complexity Problem

Standard RNA-seq analysis measures *whether* transcript abundance changes between conditions. But a critical complementary question remains underexplored: *how* does the diversity of isoforms change? A gene may show little change in total abundance while dramatically reshuffling its isoform repertoire—a phenomenon that current methods largely miss.

Entropy quantifies precisely this: the complexity, richness, and balance of isoform heterogeneity. By measuring entropy across different biological scales (the parameter q introduced above), researchers can detect whether changes are driven by shifts in rare isoforms (low q) or reorganization of dominant variants (high q). The beauty of the Tsallis framework is that you obtain a complete picture of isoform heterogeneity by computing across a range of entropic indices—a “ q -curve”—revealing which aspects of isoform organization change between conditions.

Mathematical Foundation and Interpretation

Tsallis Entropy: Definition and Intuition

For a discrete probability vector $p = (p_1, \dots, p_n)$ representing isoform proportions within a gene, Tsallis entropy is defined as:

$$S_q(p) = \frac{1 - \sum_{i=1}^n p_i^q}{q - 1}.$$

This parametric family unites diverse entropy concepts under a single framework:

- **Generalization:** Extends beyond Shannon entropy to capture scale-dependent phenomena
- **Mathematical elegance:** Reduces to well-known diversity indices at specific entropic indices
- **Practical flexibility:** Enables data-driven exploration across the full diversity spectrum

Information-Theoretic Meaning

From an information theory perspective (Shannon 1948; Furuichi 2006), entropy measures the uncertainty or surprise when drawing a single transcript from an isoform distribution. Higher entropy means the draw is less predictable (many similarly abundant isoforms), while lower entropy means one or a few isoforms dominate. This distinction is crucial: two genes with identical total abundance may have dramatically different isoform complexity.

The **q parameter acts as a sensitivity dial** that controls which aspects of the distribution become visible:

- **$q < 1$** (e.g., 0.5): Emphasizes rare, low-abundance isoforms; useful for discovering cryptic or condition-specific variants.
- **$q = 1$:** Recovers Shannon entropy; provides balanced sensitivity across all abundance scales.
- **$q > 1$:** Emphasizes dominant, abundant isoforms; captures core expression architecture.

Biological Potentiality: Why TSENAT Matters

Tsallis entropy is grounded in Shannon’s foundational information theory (Shannon 1948), generalized by Renyi’s family of entropies (Jost 2006), and extended by Tsallis (Furuichi 2006). The key innovation is a tunable parameter **q** that controls sensitivity to different aspects of the diversity distribution—which aspects of isoform heterogeneity become visible depends on the entropic index chosen (Anastasiadis 2012; Ramírez-Reyes et al. 2016; Alomani and Kayid 2023).

As stated explicitly in the mathematical literature: “This introduces the formal possibility not to set rare and common events on the same footing, as in BG or Shannon statistics, but it enhances or depresses them according to the parameter chosen.” This principle is formalized in the Hill numbers framework (Chao et al. 2010), which unifies diverse entropy-based measures (richness, Shannon, Simpson) as different manifestations of the same parametric family with varying sensitivity to abundance scales.

The flexibility is crucial for isoform analysis. The concept of “true diversity” (Chao et al. 2010) emphasizes that diversity can be decomposed into two independent components: species richness (how many distinct isoforms exist) and evenness (how evenly distributed they are across the population). Two genes can have identical Shannon entropy yet differ dramatically in isoform structure: one might be dominated by a single abundant isoform (maximizing apparent heterogeneity at low **q** where rare variants matter), while another distributes transcripts across many isoforms equally (maximizing heterogeneity across all **q** values).

Applications and Evidence: Why Multi-Scale Entropy Matters

The multi-scale nature of Tsallis entropy makes it suited for exploring isoform complexity across diverse biological contexts. By measuring information content at different entropic indices, researchers can detect patterns invisible to traditional transcript abundance measures alone. The recent emphasis on information-theoretic approaches in computational biology Bajić (2024) reflects broader recognition that complex biological systems encode information across multiple organizational scales.

Biological Contexts Where Scale-Dependent Analysis Reveals Hidden Complexity

Isoform complexity as a biological signal: Isoform switching—reorganization of the isoform landscape without necessarily changing total gene abundance—reflects strategic shifts in protein function driven by splicing regulation. Evidence from single-cell transcriptomics demonstrates that transcript-level complexity varies systematically across cell types and developmental states (Cao et al. 2017), validating that isoform heterogeneity is a genuine biological phenomenon rather than noise. Different biological processes prioritize different organizational scales (Tarabichi et al. 2013):

- Changes in rare isoform usage (revealed through low q sensitivity) might reflect exploratory or error-correction mechanisms
- Shifts in dominant isoform selection (revealed through high q sensitivity) might reflect functional specialization or robustness demands
- The full q -curve reveals whether cellular transitions involve wholesale reorganization or targeted adjustments

TSENAT enables detection of these changes through entropy-based approaches, which capture whether complexity is increasing (diversity spreading across isoforms) or decreasing (consolidation onto dominant isoforms).

Beyond classical abundance measures: Traditional RNA-seq analysis focuses on fold-changes and differential abundance. Since isoform reorganization can occur independently of total abundance changes, entropy-based approaches complement classical methods by detecting complexity shifts that transcript-level statistics alone cannot reveal.

Mechanistic Evidence from Disease and Evolution

Peer-reviewed literature provides strong empirical support for entropy-based analysis in biological contexts:

Entropy and cancer heterogeneity: Cancer cells accumulate genetic and epigenetic perturbations that systematically increase disorder in gene regulatory networks. Tarabichi and colleagues demonstrate that “Increased entropy of signaling (or gene interaction networks) has been well studied as a cancer characteristic: Network entropy increases along with cancer progresses” (Tarabichi et al. 2013).

Nijman’s complementary analysis reveals the mechanism: “cancer-associated perturbations collectively disrupt normal gene regulatory networks by increasing their entropy. Importantly, in this model both somatic driver and passenger alterations contribute to ‘perturbation-driven entropy’, thereby increasing phenotypic heterogeneity and evolvability” (Nijman 2020). This framework elegantly explains observed cancer heterogeneity without requiring that every genetic change confers an advantage—some mutations contribute entropy directly through network disruption. Increased entropy in gene regulatory networks thus drives phenotypic heterogeneity and cellular plasticity, suggesting that transcript-level entropy captures similar organizational principles (Nijman 2020).

TSENAT Main Workflow

We now demonstrate a complete workflow for detecting **isoform switching**—when cells reorganize their isoform landscape without necessarily changing total gene abundance. This often reflects strategic shifts in protein function driven by splicing regulation.

In this workflow, you will:

1. Load transcript counts and sample metadata

2. Identify genes with significant scale-dependent isoform complexity changes (using linear model interaction testing)
3. Assess which individual transcripts drive those changes (using jackknife resampling (Efron and Tibshirani 1993))
4. Interpret results in the context of paired experimental designs

We'll work with a paired experimental design where treated and control samples are linked, enabling detection of robust biological signals while controlling for subject-level variability. By the end, you'll know not just *which genes* undergo isoform switching, but *which transcripts* are responsible and *at which diversity scales* the switching occurs.

Load data and metadata

An example dataset is included for demonstration.

```
# Load packages
suppressPackageStartupMessages({
  library(TSENAT)
  library(ggplot2)
  library(SummarizedExperiment)
})

# Setup random seed for reproducibility
set.seed(42)
```

Now we will load the example dataset and associated metadata:

```
# Load example dataset (lazy-loaded as 'readcounts' by default)
# This includes SALMON preprocessing outputs:
# - readcounts: transcript-level NumReads (raw fragment counts)
# - tpm: TPM matrix (length- and library-normalized estimates)
# - effective_length: EffectiveLength vector (read-length corrected)
data(readcounts)

readcounts <- as.matrix(readcounts)

metadata_df <- read.table(
  system.file("extdata", "metadata.tsv", package = "TSENAT"),
  header = TRUE, sep = "\t"
)

gff3_file <- system.file("extdata", "annotation.gff3.gz", package = "TSENAT")
```

Data preprocessing and filtering

Next, we will create a `TSENATAnalysis` object containing a `SummarizedExperiment` data container (a standard Bioconductor class for storing high-throughput assay data along with associated metadata). You can read more about it in the Bioconductor documentation.

We will use the GFF3.gz annotation file for extracting the transcript-to-gene mapping and building the analysis object. We'll also include sample metadata for improved analysis. Following Bioconductor best practices, we create the configuration `FIRST`, then pass it to the builder function (fail-fast principle):

```

## Configure analysis parameters first (best practice: fail-fast principle)
## This validates all parameters before object creation
config <- TSENAT_config(
  sample_col = "sample",
  condition_col = "condition",
  subject_col = "paired_samples",
  q = seq(0, 2, by = 0.05),
  nthreads = 2,
  paired = TRUE,
  control = "normal"
)

## Build a complete `TSENATAnalysis` object
analysis <- build_analysis(
  config = config,
  readcounts = readcounts,
  metadata = metadata_df,
  tx2gene = gff3_file,
  tpm = tpm,
  effective_length = effective_length
)

```

The `TSENATAnalysis` object contains a `SummarizedExperiment` with transcript-level counts and gene annotations extracted from the GFF3 file.

To reduce noise and improve statistical power, we should filter out lowly-expressed transcripts. Expression estimates of transcript isoforms with zero or low expression might be highly variable (Jose and Lal 2013; Chakraborty 2019). For more details on the effect of transcript isoform prefiltering on differential transcript usage, see this paper.

```

# Filter lowly-expressed transcripts
analysis <- filter_analysis(analysis, stringency = "medium")

```

The `stringency = "medium"` parameter keeps transcripts present in $\geq 50\%$ of samples with TPM values above the median of mean gene TPM. This balances noise reduction with preservation of isoform diversity needed for meaningful entropy calculations.

Compute Tsallis Entropy

We now compute Tsallis entropy across your configured q -spectrum. Diversity measures provide a comprehensive framework for assessing isoform heterogeneity at multiple scales (Ramírez-Reyes et al. 2016; Gandrillon et al. 2021). We normalize entropy to $[0,1]$ range (`norm = TRUE`) for comparable cross- q assessment.

```

# Set random seed for reproducible bootstrap CI calculations
set.seed(12345)

# Compute diversity
analysis <- calculate_diversity(
  analysis,
  norm = TRUE
)

```

```
# Extract and display diversity results
diversity <- results(analysis,
  type = "diversity",
  n_genes = 4,
  sample = "SRR14800481")

print(diversity)
```

Table 1: **Table 1:** Tsallis entropy for first 4 genes across three diversity scales (sample: SRR14800481)

Gene	q=0.0	q=0.5	q=1.0	q=1.5	q=2.0
FOXJ2	0.66667	0.53796	0.48150	0.46975	0.48518
TMEM38A	1.00000	0.82704	0.72526	0.66965	0.64401
CCNE1	0.33333	0.32256	0.32749	0.34571	0.37418
MCOLN1	1.00000	0.39297	0.21567	0.16305	0.15195

With the q-spectrum we can produce a q-curve per sample and gene. These curves show how diversity emphasis shifts from rare to dominant isoforms as q increases and form the basis for interaction tests. The q-curve shows entropy as a function of q . Diverging curves between groups indicate scale-dependent diversity differences: separation at low q implies differences in rare isoforms, while separation at high q signals differences in dominant isoforms.

```
# Plot overall q-curve
p_qcurve <- plot_diversity_spectrum(analysis)
print(p_qcurve)
```

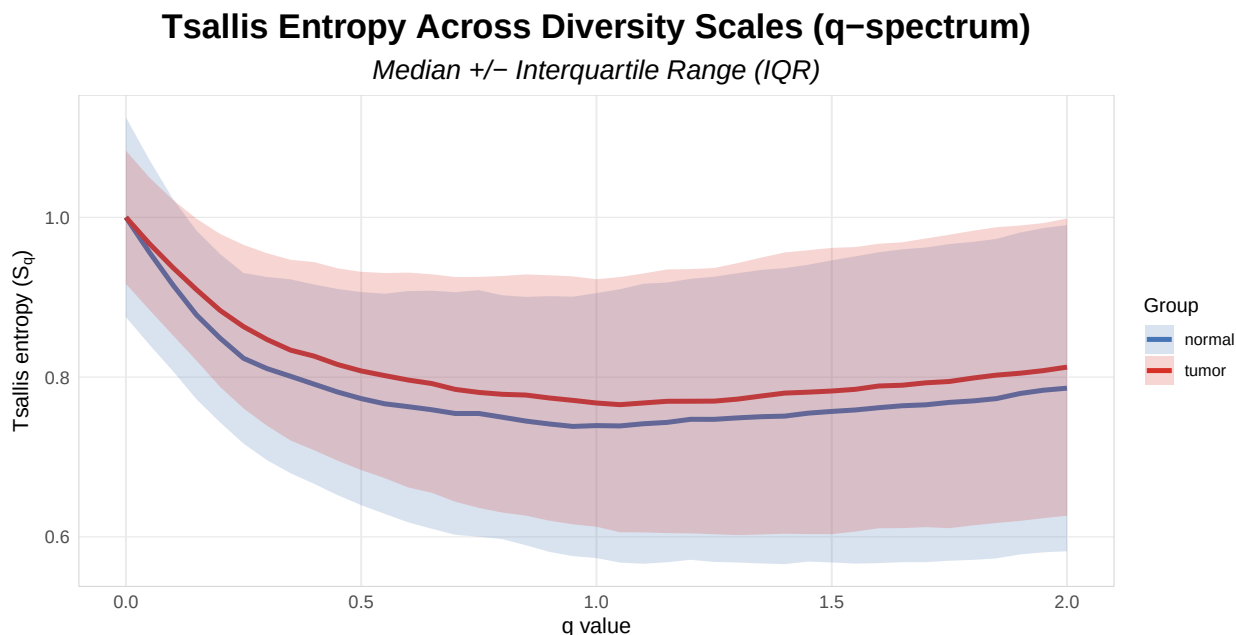


Figure 1: **Figure 1:** Isoform diversity profiles across entropic indices. Lines show normalized Tsallis entropy (0-1) for each sample, blue control and red treatment.

Quality Control: Sample Influence Assessment

Before proceeding with group-level comparisons, we should assess whether individual samples exert disproportionate influence on our entropy estimates (Efron, Bradley and Tibshirani, Robert J. 1993). To address this, we employ a leave-one-out influence assessment combined with robust M-estimation (Ernst 2004) (iteratively re-weighted least squares with Huber loss). This approach quantifies how much each sample's removal affects the estimated location differences across the q-spectrum, providing a sample-level quality control metric independent of group assignment.

Samples with high influence scores exert disproportionate leverage on entropy estimates and may warrant deeper investigation for technical quality issues, subject-specific outliers, or genuine biological signals that require careful interpretation.

We configure three key parameters: `loss_type = "huber"` employs Huber M-estimation (robust to outliers with bounded influence), `q_combine_method = "mean"` averages influence metrics across the entire q-spectrum for a single score per sample, and `influence_threshold = 0.75` designates the percentile cut-off above which samples are flagged (such that the top 25% most influential samples are reviewed).

```
# Perform multi-q sample influence analysis via S4 wrapper
analysis <- calculate_m_estimator(
  analysis,
  loss_type = "huber",
  q_combine_method = "mean",
  influence_threshold = 0.75
)

# Extract results from metadata using accessor function
sample_qc <- metadata(analysis, "m_estimate_results")
print(sample_qc)
```

Table 2: **Table 2 | Sample Influence Assessment via M-estimation.** Ranked by magnitude of resampling-based influence values. Columns: Sample identifier; condition; proportion affected; distance from centroid; outlier status. M-estimation identifies influential samples for sensitivity analysis.

Sample	Condition	Proportion_Affected	Distance_from_Centroid	Status
SRR14800481	normal	0.847	1.7930	OK
SRR14800480	normal	0.806	1.2384	OK
SRR14800477	normal	0.833	1.5809	OK
SRR14800476	normal	0.889	0.9118	OK
SRR14800475	normal	0.861	1.6796	OK
SRR14800490	normal	0.849	1.3522	OK
SRR14800489	normal	0.875	1.0969	OK
SRR14800488	normal	0.861	1.6536	OK
SRR14800479	tumor	0.861	2.2114	OK
SRR14800478	tumor	0.903	1.5721	Flag for QC
SRR14800487	tumor	0.861	1.2201	OK
SRR14800486	tumor	0.903	1.4798	Flag for QC
SRR14800485	tumor	0.861	1.4140	OK
SRR14800484	tumor	0.889	1.1499	OK
SRR14800483	tumor	0.861	1.6020	OK
SRR14800482	tumor	0.917	1.6711	Flag for QC

Interpretation: Samples with distance from centroid >1.5 or proportion of affected genes >0.85 are flagged for quality control review. Flagged samples may reflect genuine biological heterogeneity or technical artifacts requiring careful examination before proceeding with downstream analysis.

Linear-Model interaction

Each gene produces a **q-curve**: a trajectory showing how entropy changes across entropic scales from rare (low q) to abundant (high q) isoforms. Our goal is to detect whether this curve differs between groups—in other words, whether the effect of q on entropy **depends on** which group a sample belongs to. This is captured statistically as a **q x condition interaction**: if significant, it reveals genes with condition-specific diversity patterns that change shape across the diversity spectrum. Linear models (and their extensions) naturally formalize this multi-scale comparison, making them ideal for identifying such scale-dependent biological signals.

We can test for these interactions using one of four methods, each with specific strengths:

- GAM (generalized additive model): flexibly captures nonlinear q -response patterns via adaptive smoothing splines. Default method.
- LMM (linear mixed models): models subject-level random intercepts to account for within-subject correlation in repeated q -ordered measurements.
- GEE (generalized estimating equations): particularly useful for paired and longitudinal designs with repeated q -measures.
- FPCA (functional principal component analysis): treats q -values as ordered functional data, implicitly capturing q -ordering structure.

All methods automatically account for the AR(1) correlation structure inherent in Tsallis entropy measurements.

```
# Linear-model interaction test across q values using S4 wrapper
analysis <- calculate_lm(
  analysis,
  method = "gam",
  multicorr = "hochberg"
)

# Extract LM interaction results
lm_results <- results(analysis, type = "lm", rankBy = "pvalue", n = 10)

print(lm_results)
```

Table 3: **Table 3 | Leading genes with scale-dependent condition effects from generalized additive models.** We display the 6 genes with lowest adjusted p-values ($q < 0.05$). Benjamini-Hochberg correction applied; columns show gene identifier, effect size, test statistic, convergence status, and heteroscedasticity detection. Methods: GAMs with q and condition as smooth predictors (Benjamini and Hochberg 1995).

Gene	P-value	Adj. P-value	Effect Size	Test Statistic	Model Converged	Heteroscedasticity
CXCL12	3.93e-164	2.99e-162	0.8127	752.5086	TRUE	TRUE
THY1	9.80e-97	7.35e-95	0.6008	442.1369	TRUE	TRUE

Gene	P-value	Adj. P-value	Effect Size	Test Statistic	Model Converged	Heteroscedasticity
ING3	2.72e-77	2.02e-75	0.3533	352.5945	TRUE	TRUE
SNHG10	9.81e-75	7.16e-73	0.0515	340.8200	TRUE	TRUE
LINC03040	1.87e-57	1.35e-55	0.7380	261.2380	TRUE	TRUE
HDAC2	2.61e-51	1.85e-49	0.2844	232.9431	TRUE	TRUE

Interpretation: TRUE in the Heteroscedasticity column indicates that the model satisfies homogeneity-of-variance assumptions across the q-spectrum; FALSE values suggest variance heterogeneity and warrant caution in result interpretation.

Now we will plot the q-curve profile for the top genes identified by the linear-model interaction test.

```
# Plot q-curve profiles for the top 4 genes using the S4 wrapper
combined_plot <- plot_lm(
  analysis,
  n_top = 4
)
print(combined_plot)
```

q-curve: Top genes with group interaction

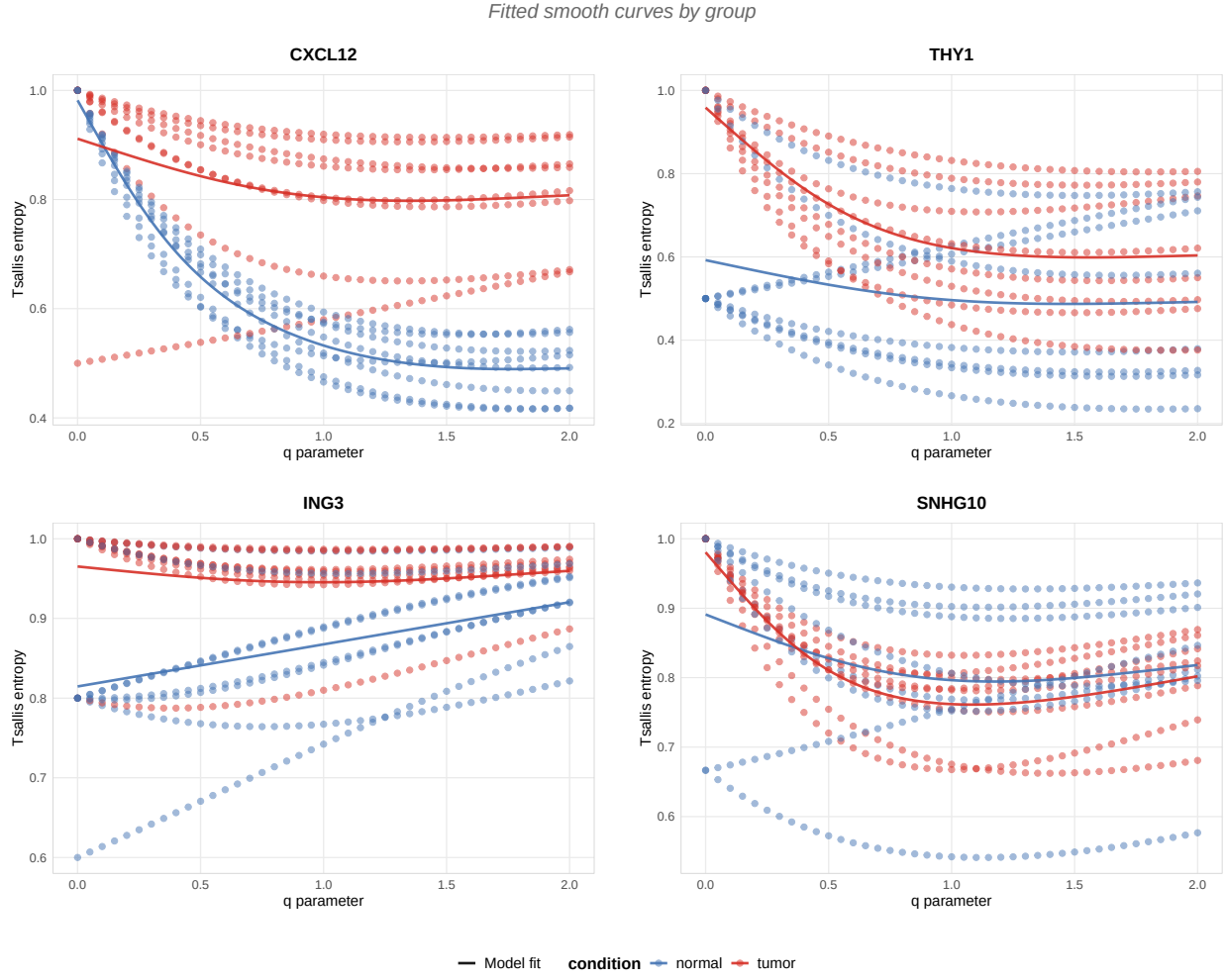


Figure 2: **Figure 2:** Scale-dependent interaction analysis. GAM-identified genes showing significant $q \times \text{condition}$ effects (Benjamini-Hochberg $q < 0.05$).

Transcript Switching Across Diversity Scales

The LM interaction tests whether condition effects depend on which diversity scale (q -value) you examine. This section identifies **which individual transcripts** drive these scale-dependent patterns, revealing whether the same transcripts switch across all scales or whether different regulatory mechanisms dominate at rare versus abundant isoform scales.

Two-Stage Analysis Approach

This workflow combines two complementary statistical methods:

1. **Multi- q LM interaction test:** Tests whether the condition effect on diversity depends on q (overall pattern across diversity scales)
2. **Single- q jackknife resampling:** Tests which individual transcripts drive those patterns at each q -value, with robustness assessment

Delta influence (from jackknife resampling (Efron, Bradley and Tibshirani, Robert J. 1993)) quantifies *how each transcript's relative importance changes between normal and tumor conditions*, weighted by stability across bootstrap iterations (positive = more influential in normal condition; negative = more influential in tumor condition).

Key Interpretation Questions

- **Consistent switching:** Do the same transcripts show significant delta influence across all q -values? Suggests robust, scale-independent splicing shift.
- **Mixed/scale-dependent switching:** Do different transcripts matter at different q -values? Suggests regulatory complexity where mechanisms differ by scale.
- **Classification:** Does isoform reorganization involve the same transcripts across scales, or a layered process where different regulatory inputs dominate at rare vs. abundant scales?

```
# Multi-Q analysis: Test switching at different q values
analysis <- calculate_jis(
  analysis,
  q = c(0, 0.5, 1, 1.5, 2),
  nboot = 100,
  lm_p_threshold = 0.05,
  threshold = 90
)
```

We configure three key parameters for the switching analysis: `nboot = 100` specifies 100 bootstrap resamples to estimate robust confidence intervals for delta influence values (Efron, Bradley and Tibshirani, Robert J. 1993); `lm_p_threshold = 0.05` filters genes to those with significant $q \times \text{condition}$ interaction effects ($p < 0.05$) before assessing transcript switching; and `threshold = 90` designates transcripts with support $\geq 90\%$ across bootstrap samples as robust switches. These parameters balance sensitivity (detecting switching) with specificity (avoiding false positives from noise).

The tables below show the transcript switching patterns for the top genes with strongest $q \times \text{condition}$ interactions, automatically computed via bootstrap resampling. The **Direction Consistency** column classifies each transcript: “Consistent” indicates stable switching across entropic indices, while “Mixed” indicates scale-dependent switching. This reveals whether isoform shifts involve the same transcripts across scales or whether different entropic indices emphasize different transcripts.

```
# Retrieve gene switching tables
tables_result <- results(analysis, type = "switching_tables")
```

Gene: CXCL12 (ENSG00000107562.18)

Transcript	q=0.50	q=1.00	q=1.50	Direction Consistency
ENST00000343575.11	0.000	0.041	0.062	Mixed directions
ENST00000374426.6	-0.063	-0.051	-0.040	Consistent negative
ENST00000374429.6	0.164	0.156	0.123	Consistent positive

Gene: THY1 (ENSG00000154096.15)

Transcript	q=0.50	q=1.00	q=1.50	Direction Consistency
ENST00000524970.5	-0.046	0.002	0.014	Mixed directions
ENST00000900758.1	0.045	0.031	0.019	Consistent positive
ENST00000956364.1	-0.075	-0.092	-0.102	Consistent negative

Delta Influence Across Diversity Scales

Visualize switching patterns for the top genes identified by the LM interaction test. This shows which transcripts are switching in genes with significant $q \times \text{condition}$ interaction effects. The heatmaps below display **jackknife delta influence** (Efron and Tibshirani 1993)-a resampling-based measure of how robustly each transcript's relative contribution changes between conditions-across different entropic indices and transcripts for each gene. Delta influence values are computed from bootstrap resampling iterations, providing robust, non-parametric estimates of transcript switching significance weighted by consistency across replicates.

```
# Generate multi-Q heatmaps for top 4 genes using S4 wrapper
plot_jis_delta(analysis, n_genes = 4)
```

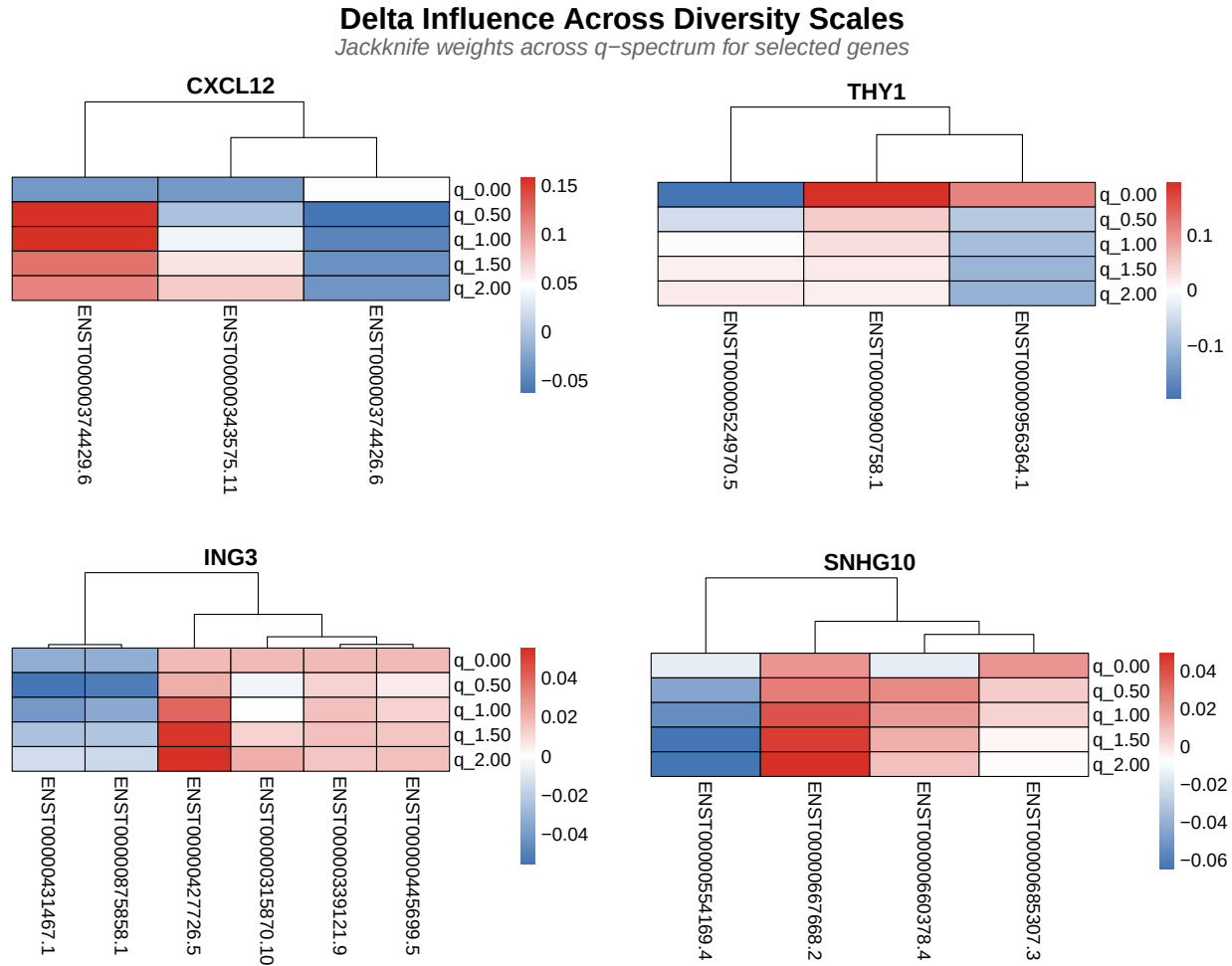


Figure 3: **Figure 3:** Transcript-level switching patterns via jackknife resampling. Rows q -values (0-2.0), columns transcripts. Red/blue indicates condition influence.

Interpretation: Rows represent q-values across the diversity spectrum (0 to 2.0, with low q emphasizing rare isoforms and high q emphasizing abundant isoforms).

- **Red cells** indicate transcripts with strong positive delta influence in the **normal condition** (first condition)—meaning these transcripts became *more important* in normal samples compared to tumor samples.
- **Blue cells** indicate negative delta influence in the normal condition (or equivalently, positive influence in the tumor condition)—meaning these transcripts became *less important* in normal samples but *more important* in tumor samples.
- **Color intensity** reflects robustness across bootstrap resamples: bright red/blue = consistent signal, pale/white = noisy or uncertain signal.

This robustness-weighted visualization reveals not just *which* transcripts switch, but *how reliably* they switch, allowing distinction between robust biological signals and artifacts of sampling variation.

Finally we can visualize the isoform expression profiles.

```
# Generate transcript abundance heatmap
plot_expression(analysis, top_n = 4, metric = "median")
```

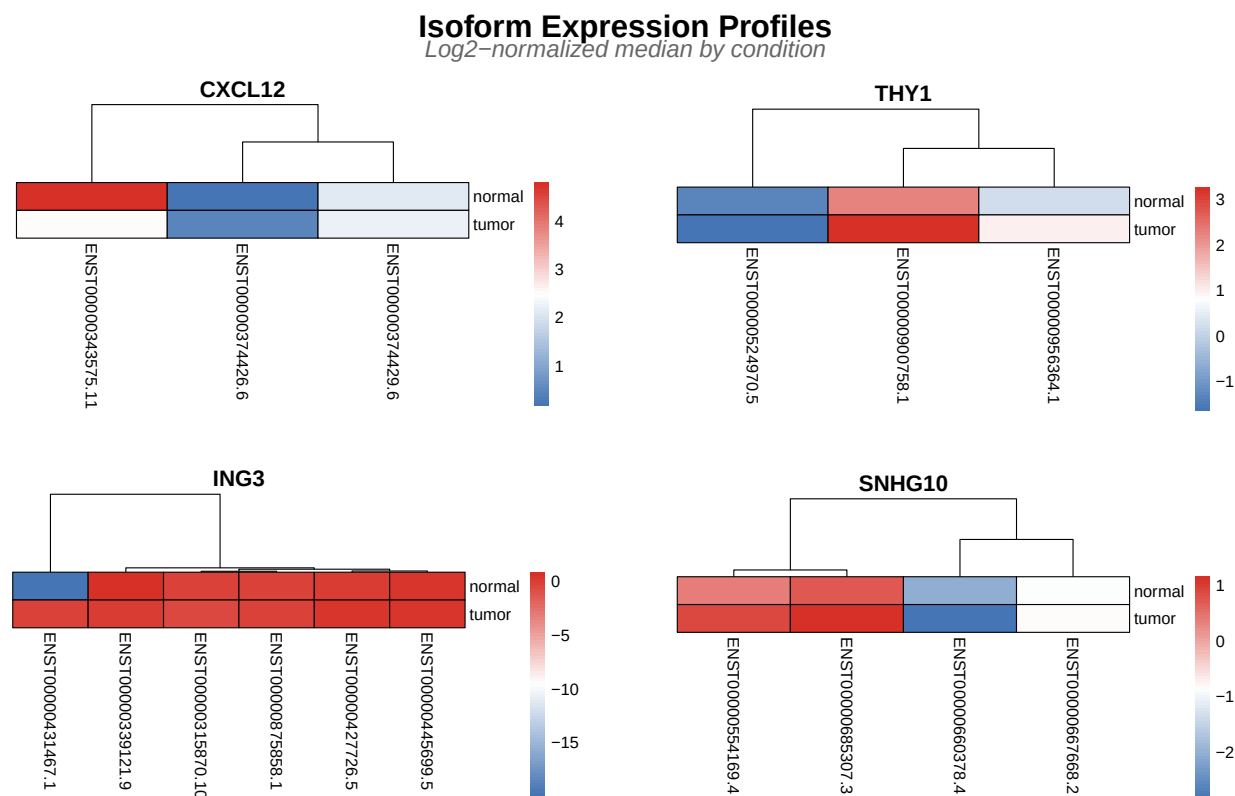


Figure 4: **Figure 4:** Transcript abundance heatmap with hierarchical clustering. Rows are transcripts, columns are conditions. Blue low, red high expression.

This heatmap reveals a critical blind spot in standard transcript quantification approaches: raw abundance measures reduce each transcript to a single number, completely obscuring the relational structure that defines isoform switching. TSENAT's entropy-based approach captures precisely what this heatmap reveals: not just which transcripts change in abundance, but how the diversity and balance of the isoform repertoire itself changes.

Effect Size Analysis

To understand which diversity scales show the most pronounced biological differences between groups, we will compute Tsallis divergence D_q across the q-spectrum (Kullback and Leibler 1951; Furuichi 2006; Jost 2006; Erven and Harremoës 2014; Sason 2022; Shiner et al. 2002)]. This metrics allows to reveal whether group differences are driven by rare isoforms (low q), dominant isoforms (high q), or uniformly across scales. Values $D > 0.1$ indicate meaningful information-theoretic separation between conditions, revealing that isoform complexity patterns fundamentally differ between treatment groups across all q-dependent scales (Ré and Azad 2014).

For two probability distributions P and Q representing isoform proportions in control and treatment conditions, Tsallis divergence is:

$$D_q(P||Q) = \frac{1 - \sum_i p_i^q \cdot q_i^{1-q}}{q - 1}$$

where p_i and q_i are the probability values at position i .

```
# Computes pairwise information-theoretic distance (Tsallis divergence)
analysis <- calculate_divergence(analysis)
```

Table 4: **Table 3 | Pairwise Tsallis divergence estimates.** Divergence (distance) between conditions from Tsallis entropy framework. Columns: gene identifier; pairwise comparison; divergence value; confidence interval (95%). Ranked by magnitude.

	q_0.01	q_0.05	q_0.1
FOXJ2	0.04377	0.04633	0.04952
TMEM38A	0.14876	0.15768	0.16891
GSR	0.08359	0.08801	0.09349
SNX4	0.06378	0.06744	0.07200

```
# Compute effect sizes
analysis <- calculate_effect_sizes(
  analysis,
  significance_threshold = 0.05,
  enrich_per_q_pattern = TRUE
)

# Extract top 6 genes by statistical significance
top_genes_result <- results(
  analysis,
  type = "effect_sizes_divergence",
  top_n = 6,
  sort_by = "p_value_interaction"
)

print(top_genes_result)
```


Table 5: Top 6 Genes by Effect Size (Tsallis Divergence)

Gene	P-value	Slope Diff	D(q=0.5)	D(q=1.0)	D(q=2.0)	Pattern
CXCL12	0	-0.3725	0.3115	0.3697	0.1643	Balanced
THY1	0	0.2928	0.1481	0.1720	0.0589	Balanced
ING3	0	0.1136	0.0122	0.0137	0.0040	Rare driven
SNHG10	0	0.1203	0.0874	0.0956	0.0283	Rare driven
LINC03040	0	0.1063	0.2850	0.3119	0.1098	Rare driven
HDAC2	0	0.1244	0.0651	0.0699	0.0201	Rare driven

Interpretation of Pattern Types:

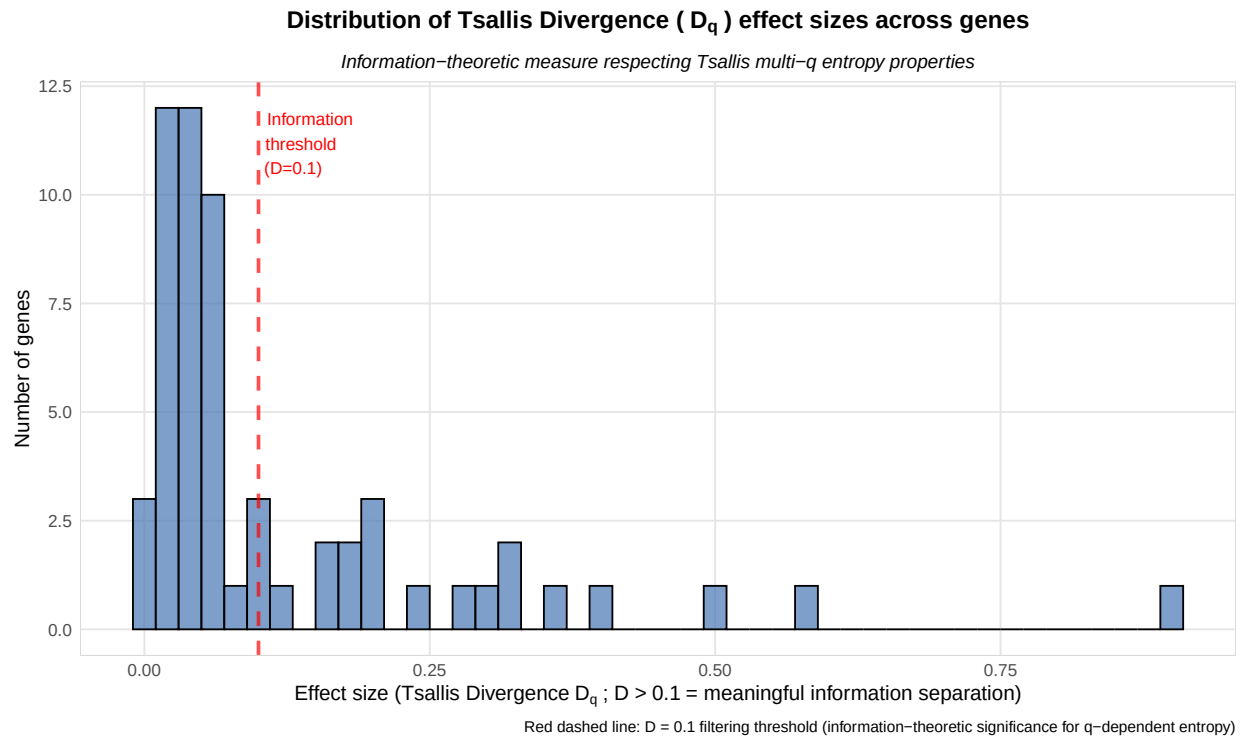
- Rare driven: $D(q=0.5) \gg D(q=2.0)$. Low-abundance isoforms are condition-specific; treatment group preferentially expresses rare transcripts not seen in control.
- Abundant driver: $D(q=0.5) \ll D(q=2.0)$. High-abundance isoforms shift between conditions; treatment remodels the dominant transcript landscape without rare variants changing.
- Balanced: $D(q=0.5) \sim D(q=2.0)$. All isoforms shift proportionally; no preferential weighting to rare or abundant forms; suggests systematic rebalancing.

The effect sizes reveal which genes show the most information-theoretic separation across the q-spectrum **between paired treatment groups**. Genes with Tsallis divergence $D > 0.1$ show substantial divergence, indicating that entropy distributions differ fundamentally between conditions across the full q-spectrum.

Biological Interpretation: Following Tsallis entropy theory and information-theoretic principles (Furuichi 2006; Jost 2006; Erven and Harremoës 2014; Hyndman and Athanasopoulos 2018), effect size quantification (Chanda et al. 2020) genes with large D values are those where isoform complexity distributions differ qualitatively between conditions when examined at all sensitivity scales ($q = \text{rare} \rightarrow \text{abundant}$ isoforms). For example, a gene might show high entropy at low q (emphasizing rare isoforms) in one condition but low entropy at high q (emphasizing abundant isoforms) in another, revealing scale-dependent regulatory mechanisms. These genes are candidates for investigation of condition-specific splicing architecture and dynamic isoform switching.

```
# Visualize the distribution of Tsallis divergence effect sizes across genes
plot_obj <- plot_divergence_distribution(
  analysis,
  threshold = 0.1
)

print(plot_obj)
```



```
# Visualize q-spectrum curves for top 4 genes by significance
p_multi <- plot_divergence_spectrum(
  analysis,
  n_genes = 4,
  use_pvalue_ranking = TRUE
)

print(p_multi)
```

Interpreting the Q-Spectrum Curve:

- **Shape matters more than single value:** The **q-spectrum** curve is the complete biological story (Chao et al. 2010). A flat curve (balanced) vs. declining curve (rare driven) vs. rising curve (abundant driven) encode fundamentally different mechanisms (Sason 2022). As described in the pattern table above, genes falling into each category reveal different regulatory strategies.
- **$q=1$ (KL divergence):** The middle point corresponds to ordinary Kullback-Leibler divergence (Kullback and Leibler 1951; Ré and Azad 2014), where the Tsallis divergence reduces to standard KL divergence in the limit $q \rightarrow 1$. This weights all isoforms equally.

We can also plot the global divergence spectrum across all genes. This curve is the full biological signature of **isoform switching**-the pattern encodes which abundance scales (rare vs. abundant) are affected.

```
# Visualize global divergence q-spectrum across all genes (aggregated)
# Plot is rendered directly by knitr (no file dependency)
p <- plot_divergence_spectrum(analysis)

print(p)
```

Statistical Validation During Interpretation:

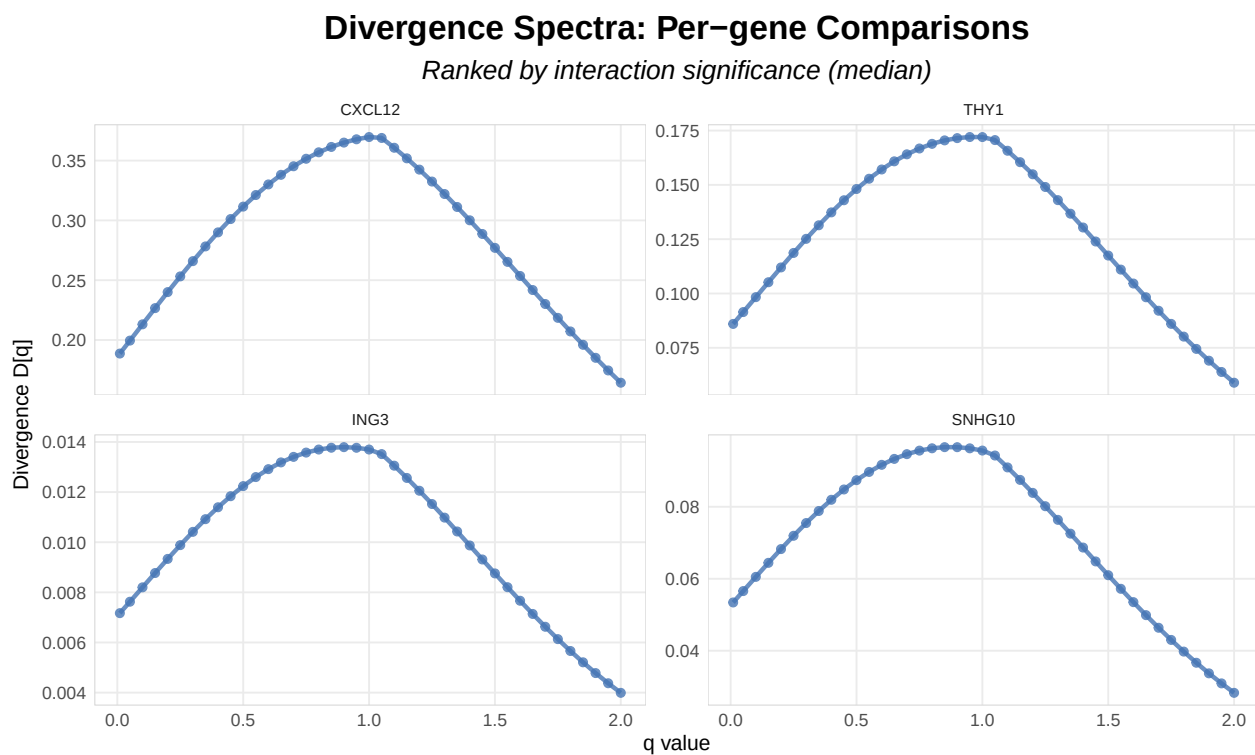


Figure 5: **Figure 5:** Q-spectrum curves for top genes by significance. Each line represents divergence as a function of q -value; shape reveals biological pattern (rare-driven, balanced, or abundant-driven).

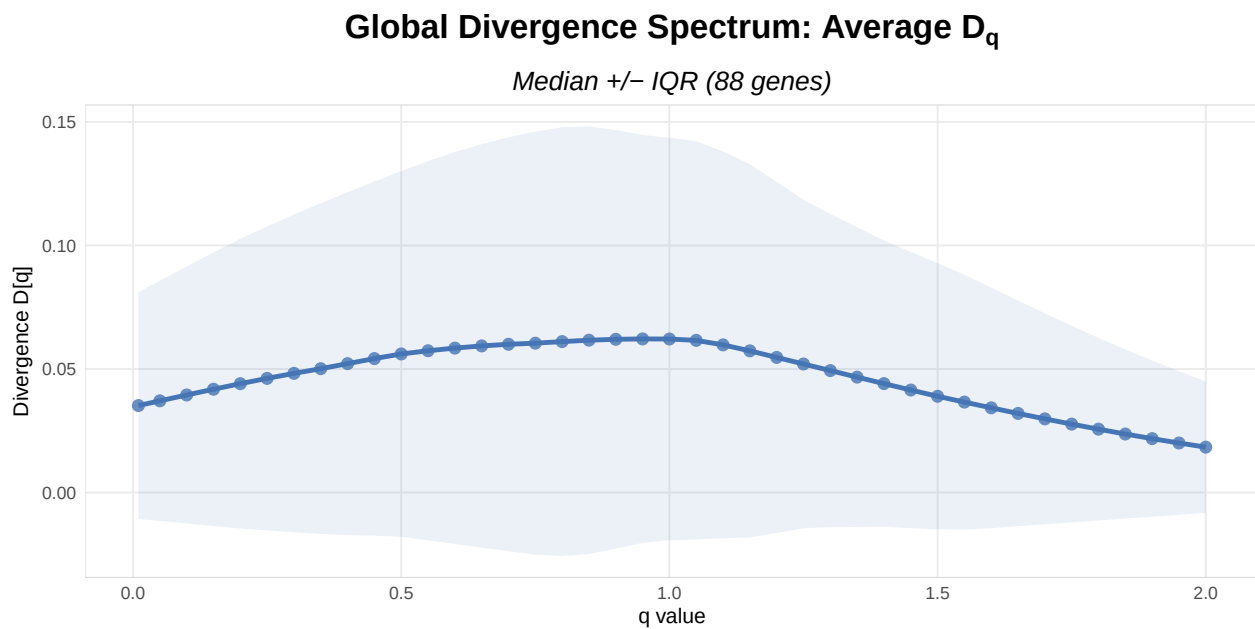


Figure 6: **Figure 6:** Global divergence spectrum across all genes. Aggregated q -spectrum pattern shows which abundance scales (rare vs. abundant isoforms) are affected genome-wide.

1. Bootstrap CI validity: Per-q CIs should narrow as q increases (higher q = aggregation effect, less variance). If CIs grow with q, check for data quality issues (Efron and Tibshirani 1993; Efron, Bradley and Tibshirani, Robert J. 1993).
2. Monotonicity check: By Tsallis entropy theory, entropy is monotone decreasing in q. Divergence should NOT show erratic increases with q. Small fluctuations are normal, but large spikes indicate numerical instability (Furuichi 2006).
3. Comparison with genome-wide patterns: Compute q-spectra for housekeeping genes (GAPDH, ACTB, etc.). These should show BALANCED patterns. If not, revisit normalization parameters (Erhard et al. 2018).

Appendices

This main vignette is complemented by two comprehensive appendices:

Appendix A: Equivalence Validation - TSENAT vs SplicingFactory

Objective: Validates that TSENAT's Shannon and Simpson entropy implementations are mathematically equivalent to SplicingFactory.

View Appendix A

Appendix B: Non-Parametric Validation of Statistical Test Results via GAM and Rank-Based Methods

Objective: Validate q-value * group interaction detection results from statistical tests using complementary non-parametric modeling approaches.

View Appendix B

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Session Information

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#> Running under: Ubuntu 22.04.5 LTS
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#> Matrix products: default
#> BLAS/LAPACK: /home/nouser/miniconda3/lib/libopenblas-r0.3.30.so; LAPACK version 3.12.0
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#>  [3] LC_TIME=de_DE.UTF-8      LC_COLLATE=es_ES.UTF-8
#>  [5] LC_MONETARY=de_DE.UTF-8  LC_MESSAGES=es_ES.UTF-8
#>  [7] LC_PAPER=de_DE.UTF-8     LC_NAME=C
#>  [9] LC_ADDRESS=C             LC_TELEPHONE=C
#> [11] LC_MEASUREMENT=de_DE.UTF-8 LC_IDENTIFICATION=C
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#> time zone: Europe/Berlin
#> tzcode source: system (glibc)
#>
#> attached base packages:
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#> [8] base
#>
```

```

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#> [3] knitr_1.51                    dplyr_1.2.0
#> [5] SummarizedExperiment_1.40.0 Biobase_2.70.0
#> [7] GenomicRanges_1.62.1        Seqinfo_1.0.0
#> [9] IRanges_2.44.0              S4Vectors_0.48.0
#> [11] BiocGenerics_0.56.0          generics_0.1.4
#> [13] MatrixGenerics_1.22.0        matrixStats_1.5.0
#> [15] kableExtra_1.4.0            TSENAT_0.99.0
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